Task 1: Personal Information

Program to display my name, age, qualification, college name and city on separate lines.

```
# Personal Information
name = "Shaik Sahedabe"
age = 22
qualification = "B.Tech CSE"
college = "NRI Institute of Technology"
city = "Ongole"

print("Name:", name)
print("Age:", age)
print("Qualification:", qualification)
print("College:", college)
print("City:", city)
```

Explanation: I stored my details in variables and used print() to display each detail on a separate line.

Task 2: Simple Calculator

Program to perform addition, subtraction, multiplication and division on two numbers.

```
# Simple Calculator
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)

if b != 0:
    print("Division =", a / b)
else:
    print("Cannot divide by zero")
```

Explanation: I used float() to get numbers from the user. The if condition checks that the second number is not zero before division.

Task 3: Area Calculator

Program to find the area of a rectangle, square and circle.

```
# Area Calculator
length = float(input("Enter rectangle length: "))
breadth = float(input("Enter rectangle breadth: "))
side = float(input("Enter square side: "))
radius = float(input("Enter circle radius: "))

rectangle = length * breadth
square = side * side
circle = 3.14 * radius * radius

print("Area of rectangle =", rectangle)
print("Area of square =", square)
print("Area of circle =", circle)
```

Explanation: Rectangle area = length × breadth. Square area = side × side. Circle area = 3.14 × radius × radius.

Task 4: Bill Generator

Program to calculate the bill for three products and display their prices.

```
# Bill Generator
product1 = "Notebook"
price1 = 50

product2 = "Pen"
price2 = 10

product3 = "Pencil"
price3 = 5

total = price1 + price2 + price3

print("Product:", product1, "Price:", price1)
```

```
print("Product:", product2, "Price:", price2)
print("Product:", product3, "Price:", price3)
print("Total bill =", total)
```

Explanation: I stored three product names and prices in variables. I added the prices to get the total bill.

Task 5: Student Result

Program to enter a student's name and marks in three subjects and calculate total and average.

```
# Student Result
name = input("Enter student name: ")

mark1 = int(input("Enter first subject marks: "))
mark2 = int(input("Enter second subject marks: "))
mark3 = int(input("Enter third subject marks: "))

total = mark1 + mark2 + mark3
average = total / 3

print("Student Name:", name)
print("Total Marks:", total)
print("Average Marks:", average)
```

Explanation: I used input() to get the name and marks. Total is the sum of three marks. Average is total divided by 3.

Task 6: Identify the Valid Identifiers

An identifier is the name given to a variable. The valid and invalid identifiers are:

```
Valid identifiers:
1. student_name1  - It starts with a letter and has an underscore and a digit.
2. student        - It contains only letters.
3. total_marks    - Underscore is allowed.
4. employee123    - Digits are allowed after the first character.
5. price          - It contains only letters.

Invalid identifiers:
1. class           - It is a Python keyword.
2. my-name         - A hyphen is not allowed in a variable name.
3. for             - It is a Python keyword.
```

Explanation: A variable name should not start with a number, contain spaces or hyphens, or use a reserved keyword.

Task 7: Code Correction

Correct program for the given code:

```
# Corrected Program
name = "Ravi"
age = 21

if age >= 18:
    print("Eligible to vote")
    print(name)
```

Explanation: I placed each statement on a separate line and used indentation inside the if block.

Task 8: Keyword Identification

Program to display Python keywords using the keyword module.

```
# Display Python keywords
import keyword

print(keyword.kwlist)

# Five keyword examples

# if is used to check a condition
age = 20
if age >= 18:
    print("Eligible")

# else is used when the if condition is false
else:
    print("Not eligible")
```

```
# for is used to repeat statements
for i in range(3):
    print(i)

# def is used to create a function
def hello():
    print("Hello")

hello()

# return is used to send a value from a function
def add(a, b):
    return a + b

print(add(5, 3))
```

Explanation: if checks a condition. else runs when the condition is false. for repeats statements. def creates a function. return sends a result back from a function.

Task 9: Employee Details

Program to store and display employee ID, name, salary and department.

```
# Employee Details
employee_id = 101
employee_name = "Ravi"
salary = 25000
department = "IT"

print("Employee ID:", employee_id)
print("Employee Name:", employee_name)
print("Salary:", salary)
print("Department:", department)
```

Explanation: I created four variables to store the employee details and displayed them using print().

Task 10: Swap Two Variables

Program to swap two numbers without using a third variable.

```
# Swap Two Variables
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before swapping:", a, b)

a, b = b, a

print("After swapping:", a, b)
```

Explanation: In Python, a, b = b, a swaps the values directly without using another variable.

Task 11: Simple Interest Calculator

Formula: Simple Interest = (Principal × Time × Rate) / 100.

```
# Simple Interest Calculator
p = 10000
t = 2
r = 5

si = (p * t * r) / 100

print("Principal =", p)
print("Time =", t)
print("Rate =", r)
print("Simple Interest =", si)
```

Explanation: I stored principal, time and rate in variables and used the given formula to calculate simple interest.

Task 12: Salary Calculator

Basic salary is Rs. 25,000. HRA is 20% and DA is 10% of basic salary.

```
# Salary Calculator
basic_salary = 25000
```

```
hra = basic_salary * 20 / 100
da = basic_salary * 10 / 100

total_salary = basic_salary + hra + da

print("Basic Salary =", basic_salary)
print("HRA =", hra)
print("DA =", da)
print("Total Salary =", total_salary)
```

Explanation: HRA is Rs. 5,000 and DA is Rs. 2,500. I added both to the basic salary. Total salary is Rs. 32,500.

How to Run the Programs

1. Open VS Code.
2. Create a new Python file, for example task1.py.
3. Type or paste the program into the file and save it.
4. Open Terminal in VS Code.
5. Run the program using: **python task1.py**
6. Enter values when the program asks for input.

Note: Tasks 4 and 9 use simple example product and employee details. You can change them if your teacher asks for different details.